

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Annexure A

Duration : 1 Hour
Total Marks:50

Design Task

Code of Conduct:

I N. Chawran Aryan (192365053) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Chawraan Aryan

Signature of the student:

Design Task

1. Hybrid AI Recommendation Engine Design

Design a hybrid AI-based recommendation system that integrates content-based and collaborative filtering techniques using PySpark RDDs. Explain how distributed data processing can be used to handle large-scale user-item interactions, and describe the role of intelligent agents in adapting recommendations based on user behavior and feedback in real time.

2. Smart Disaster Detection System

Develop a distributed AI system using PySpark and RDD operations to detect natural disasters (such as floods or earthquakes) from multi-source data streams (satellite images, sensors, and social media). Describe the system architecture, data fusion techniques, and how intelligent agents coordinate to provide early warnings and response strategies.

3. AI-Based Fraud Detection Framework

Design a scalable fraud detection system using PySpark RDDs to process large volumes of financial transactions. Explain how anomaly detection algorithms can be implemented in a distributed manner and how intelligent agents collaborate to identify suspicious patterns, trigger alerts, and continuously improve detection accuracy.

4. Autonomous Supply Chain Optimization System

Construct a distributed AI system using PySpark for optimizing supply chain operations (inventory, logistics, demand forecasting). Describe how RDD transformations support large-scale data processing and how multiple intelligent agents interact to make decentralized decisions for cost minimization and efficiency improvement.

5. Personalized Learning Analytics Platform

Design an AI-driven educational platform that uses PySpark and RDDs to analyze student learning behavior and performance data. Explain how the system supports adaptive learning through intelligent agents, real-time feedback, and scalable analytics to personalize content delivery for thousands of learners.

RUBRICS FOR CALCULATION

Evaluation Criteria	Problem Understanding & Requirement Analysis	System Architecture Design	Use of PySpark & RDD Operations	Intelligent Agent Integration	Scalability & Distributed Computing Design	Data Flow & Processing Pipeline	Innovation & Creativity	Clarity, Presentation & Documentation
Weightage	10%	20%	15%	15%	10%	10%	10%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
Problem Understanding & Requirement Analysis	Clearly defines problem, objectives, and constraints	Mostly clear with minor gaps	Basic understanding	Unclear or incorrect
System Architecture Design	Complete, scalable, well-structured architecture with diagrams	Good design with minor issues	Basic design, lacks detail	Poor/incomplete
Use of PySpark & RDD Operations	Efficient and correct use of RDD transformations/actions	Mostly correct usage	Limited or partial use	Incorrect/missing
Intelligent Agent Integration	Strong integration with clear agent roles and logic	Good integration	Basic integration	Poor/no integration

Scalability & Distributed Computing Design	Clearly addresses scalability, fault tolerance, parallelism	Covers most aspects	Limited consideration	Not addressed
Data Flow & Processing Pipeline	Clear, logical, and well-defined data flow	Mostly clear	Some gaps	Poorly defined
Innovation & Creativity	Highly innovative, realistic, and impactful solution	Some innovation	Basic idea	No originality
Clarity, Presentation & Documentation	Very clear, neat diagrams, well-organized report	Mostly clear	Somewhat organized	Poor presentation

1. Hybrid AI Recommendation Engine Design

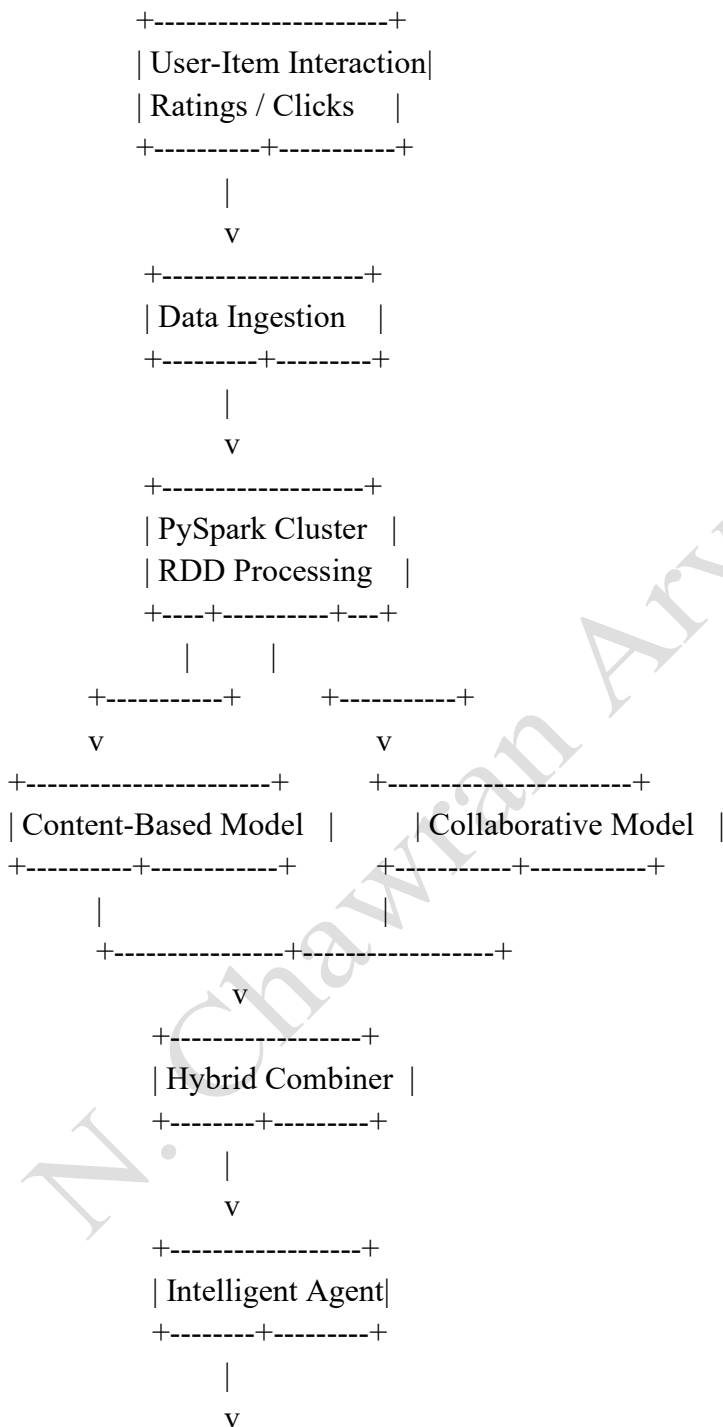
Objective

The system provides personalized recommendations by combining:

- **Content-Based Filtering** – recommends items similar to those previously liked by a user.
- **Collaborative Filtering** – recommends items based on the preferences of similar users.

This hybrid approach improves recommendation accuracy and reduces problems such as cold-start and sparse data.

System Architecture



Recommendations

Use of PySpark RDDs

Large user-item datasets are divided into partitions and processed in parallel.

Example operations:

- `map()` – convert raw records into (user, item) pairs.
- `filter()` – remove invalid ratings.
- `groupByKey()` – group interactions by user.
- `join()` – combine user behavior and item information.
- `reduceByKey()` – calculate aggregated ratings.
- `cache()` – store frequently used RDDs in memory.
- `collect()` – retrieve final recommendations.

Intelligent Agent Integration

Different agents perform specific tasks:

1. **User Behavior Agent** – monitors clicks, searches, ratings, and purchases.
2. **Recommendation Agent** – generates personalized recommendations.
3. **Feedback Agent** – collects positive and negative feedback.
4. **Adaptation Agent** – updates recommendation weights in real time.

Scalability

PySpark distributes millions of interactions across multiple worker nodes. RDD partitioning, parallel execution, caching, and fault tolerance allow the system to process large-scale data efficiently.

Conclusion

The hybrid recommendation engine combines content-based and collaborative filtering with PySpark RDD operations. Intelligent agents continuously analyze user behavior and feedback to provide accurate and adaptive recommendations.

2. Smart Disaster Detection System

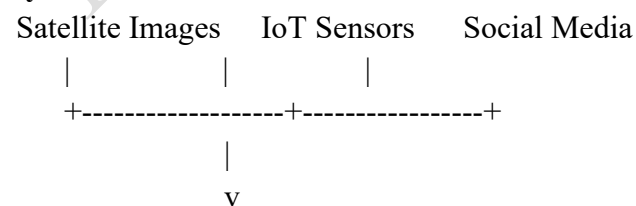
Objective

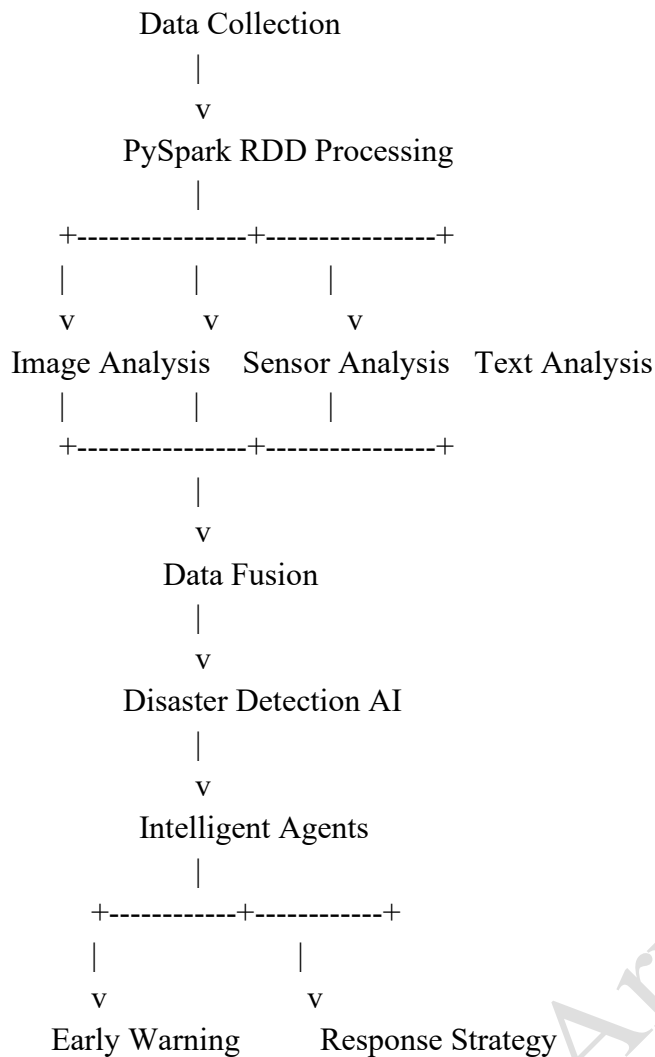
The system detects disasters such as:

- Floods
- Earthquakes
- Cyclones
- Landslides

It processes data from satellite images, IoT sensors, weather systems, and social media.

System Architecture





Data Fusion

The system combines multiple sources:

- Satellite images identify flooded or damaged regions.
- Sensors detect water levels, seismic activity, and temperature.
- Social media provides real-time reports from people.

PySpark RDD operations such as map(), filter(), union(), and reduceByKey() process and combine these datasets.

Intelligent Agents

- **Monitoring Agent** – continuously observes incoming data.
- **Detection Agent** – identifies possible disasters.
- **Verification Agent** – compares information from multiple sources.
- **Alert Agent** – sends warnings to authorities and citizens.
- **Response Agent** – recommends evacuation routes and emergency actions.

Scalability and Distributed Computing

Different data streams are distributed across worker nodes. RDDs enable parallel processing and fault recovery. If one worker node fails, lost partitions can be recomputed.

Conclusion

The distributed disaster detection system combines multi-source data, PySpark RDD processing, AI-based

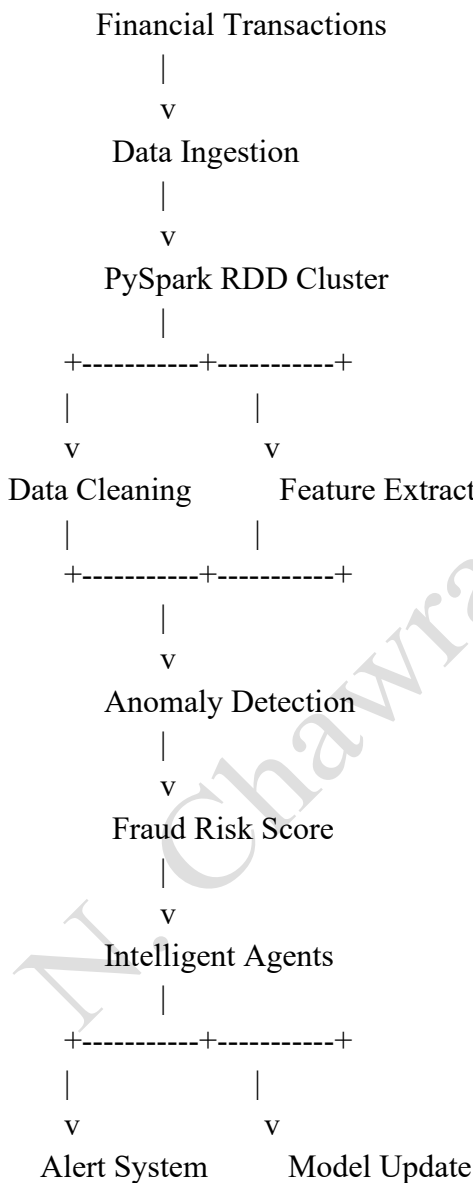
detection, and intelligent agents to provide early warnings and improve emergency response.

3. AI-Based Fraud Detection Framework

Objective

The objective is to identify suspicious or fraudulent financial transactions from a large volume of transaction data.

System Architecture



RDD Operations

Transaction records can be processed using:

map() → Extract transaction features
filter() → Remove invalid transactions
reduceByKey() → Calculate transaction statistics
groupByKey() → Group transactions by account
join() → Combine customer and transaction data
cache() → Improve repeated processing

Distributed Anomaly Detection

Each transaction is analyzed using features such as:

- Transaction amount
- Location
- Time
- Transaction frequency
- Device information
- Previous transaction behavior

An anomaly score can be calculated as:

Fraud Score = Unusual Amount
+ Unusual Location
+ High Transaction Frequency
+ Abnormal User Behavior

Transactions with scores above a threshold are marked as suspicious.

Intelligent Agents

1. **Transaction Monitoring Agent** – monitors transactions.
2. **Anomaly Detection Agent** – detects unusual patterns.
3. **Investigation Agent** – verifies suspicious activity.
4. **Alert Agent** – sends notifications.
5. **Learning Agent** – uses feedback from confirmed fraud cases to improve detection.

Scalability

RDD partitions allow millions of transactions to be processed simultaneously across multiple machines. This improves processing speed, fault tolerance, and system reliability.

Conclusion

The framework uses distributed PySpark processing, anomaly detection, and intelligent agents to identify fraud, generate alerts, and continuously improve detection accuracy.

4. Autonomous Supply Chain Optimization System

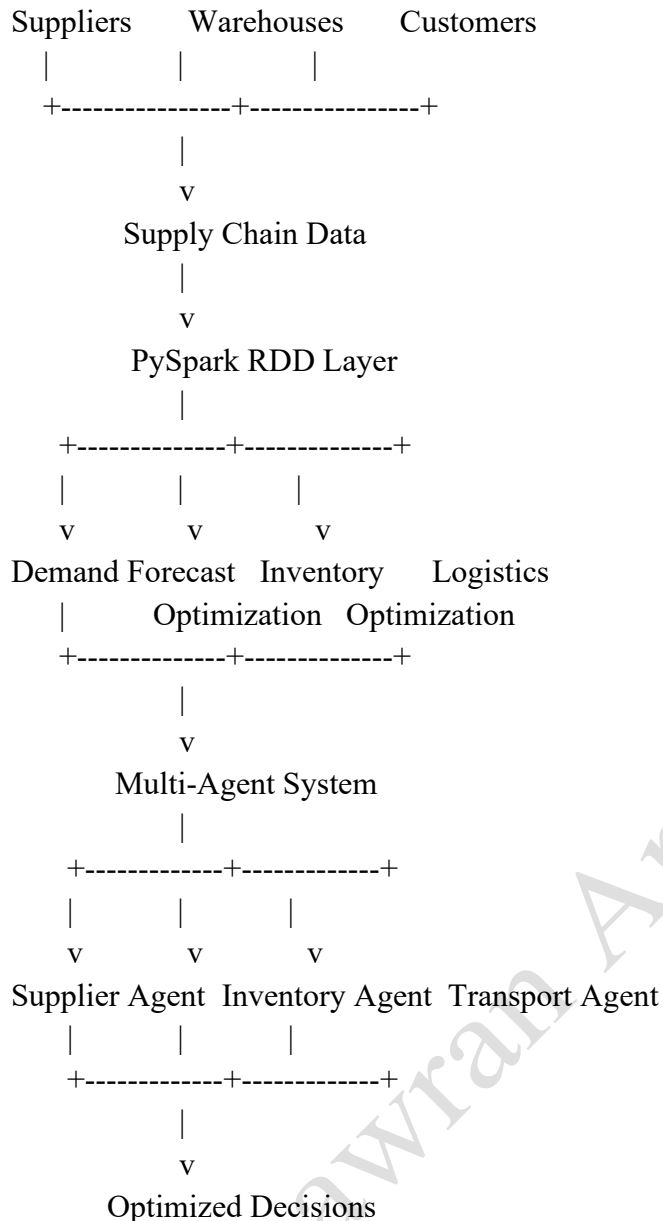
Objective

The system optimizes:

- Inventory management
- Transportation and logistics
- Demand forecasting

- Warehouse operations
- Cost minimization

System Architecture



Use of RDD Transformations

- map() – transforms raw sales data into usable records.
- filter() – removes incorrect or incomplete records.
- reduceByKey() – calculates total demand.
- groupByKey() – groups products by warehouse.
- join() – combines inventory and demand data.
- sortBy() – prioritizes delivery routes.

Intelligent Agents

1. Supplier Agent

Monitors supplier availability and delivery schedules.

2. Inventory Agent

Controls stock levels and automatically identifies reorder requirements.

3. Demand Forecasting Agent

Predicts future product demand using historical data.

4. Logistics Agent

Selects efficient transportation routes.

5. Coordination Agent

Combines decisions from all agents to minimize cost and improve efficiency.

Distributed Decision Making

Each agent can make local decisions while sharing important information with other agents. For example:

High Demand Detected



Demand Agent informs Inventory Agent



Inventory Agent checks stock



Supplier Agent arranges products



Logistics Agent plans delivery

Scalability

PySpark processes supply chain data from multiple warehouses and regions in parallel. The distributed system supports large datasets and provides fault tolerance.

Conclusion

The system integrates PySpark RDD processing with multiple intelligent agents to optimize inventory, demand forecasting, logistics, and overall supply chain efficiency.

5. Personalized Learning Analytics Platform

Objective

The platform analyzes student learning behavior and performance to provide personalized educational content for thousands of learners.

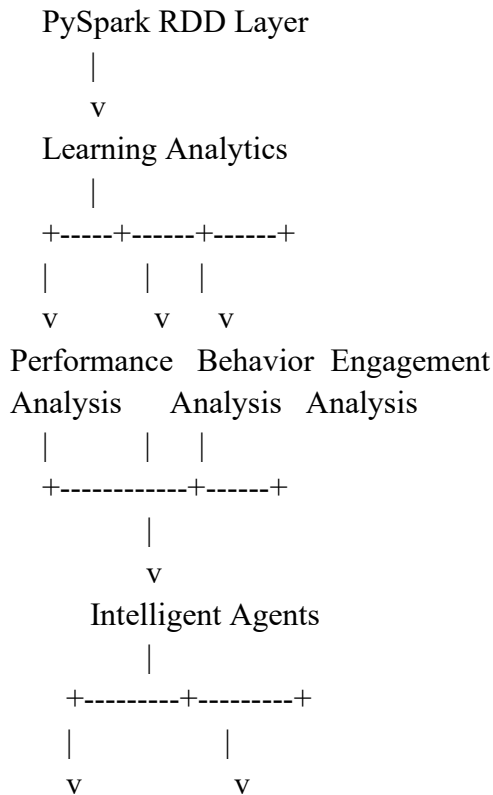
System Architecture

Student Data Sources



Data Collection





Personalized Content Real-Time Feedback

PySpark and RDD Operations

Student records are processed using:

- `map()` – extracts student performance features.
- `filter()` – selects students based on performance criteria.
- `reduceByKey()` – calculates average marks.
- `groupByKey()` – groups student activities.
- `join()` – combines attendance, quiz, and assignment data.
- `cache()` – stores frequently used analytics data.

Example:

(student_id, score)

|
reduceByKey()

(student_id, total_score)

Intelligent Agents

1. Student Profile Agent

Maintains information about each student's learning behavior and performance.

2. Learning Analysis Agent

Analyzes quiz scores, attendance, and engagement.

3. Content Recommendation Agent

Recommends suitable lessons, videos, and exercises.

4. Feedback Agent

Provides immediate feedback after assessments.

5. Adaptive Learning Agent

Changes content difficulty based on student performance.

Adaptive Learning Process

Student Activity



Data Collection



PySpark Analytics



Performance Analysis



Intelligent Agent Decision



Personalized Learning Content



Student Feedback



Continuous Improvement

Scalability

PySpark distributes student data across multiple worker nodes. Therefore, the platform can support thousands of learners simultaneously. RDDs provide parallel processing, caching, and fault tolerance.

Conclusion

The Personalized Learning Analytics Platform uses PySpark and RDD operations to process large-scale student data. Intelligent agents provide adaptive content, real-time feedback, and personalized learning experiences.